Reference Coordinate Systems (1/3)

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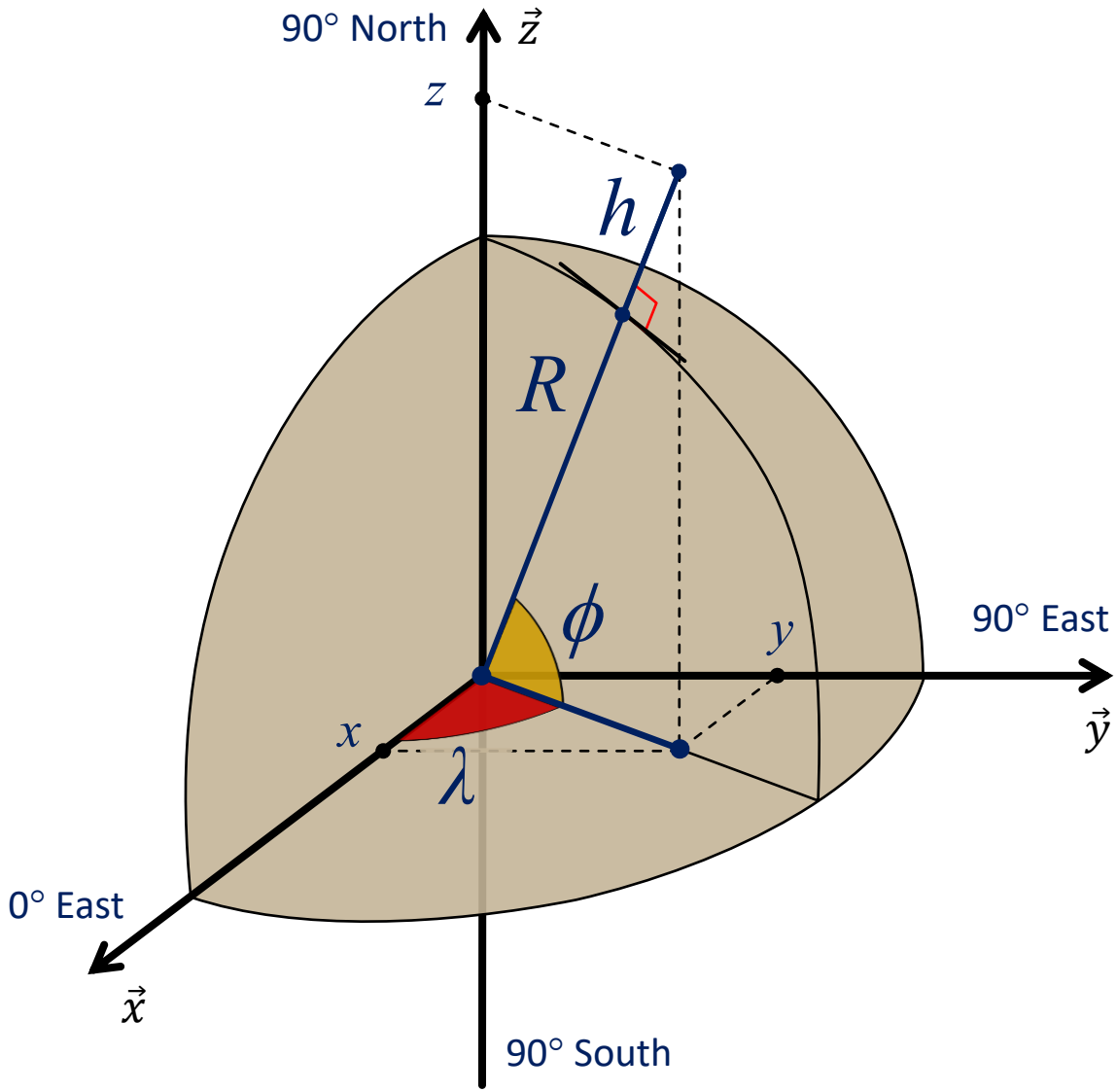
Reference Coordinate Systems

- ECEF Coordinate System
- WGS-84 and PZ-90 Frames
- Geodetic Coordinates
- Geodetic to Cartesian Coordinates
- Cartesian to Geodetic Coordinates
- The Geoid
- Reference Ellipsoids
- Local Geodetic Datums
- Datum Transformations
- Molodensky Transformation
- Rotation Matrices
- ENU Coordinates
- Azimuth & Elevation
- Orthodromes
- Loxodromes

ECEF – Earth-Centered, Earth-Fixed
WGS – World Geodetic System
PZ – Parametry Zemli (Russian)
ENU – East-North-Up

LLH on a Spherical World



ϕ - Latitude

Geocentric Latitude

λ - Longitude

h - Height

Latitude and Longitude Units/Formats

Latitude

N

S

±dd.dddddd°
dd.dddddd°N/S
dd°mm.mmmm'N/S
dd°mm'ss.ss"N/S
±r.rrrrrrr rad

Longitude

E

W

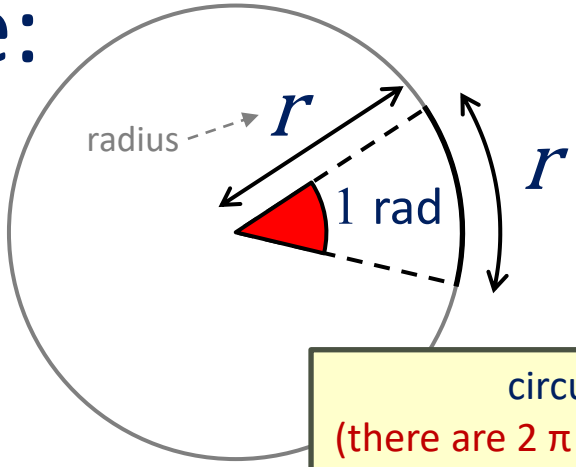
±ddd.dddddd°
ddd.dddddd°E/W
ddd°mm.mmmm'E/W
ddd°mm'ss.ss"E/W
±r.rrrrrrr rad

$1° = 60'$ (arc minutes)
$1' = 60''$ (arc seconds)
$180° = \pi\ rad$

1 m precision

Radians (rad)

circle:



$\pi = 3.14159\ 26535\ 8979\ 32384\ 62643$

$1\ rad = 57°.2957\ 79513\ 08232\ 08767\ 98155$

$1\ rad = 206264''.80624\ 70963\ 55156$

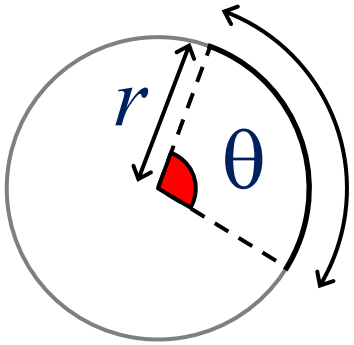
circumference = 2 π r

(there are 2 π radians in a circumference)

Circular arc length:

$$l = r\ \theta_{[rad]}$$

same units



l (arc length)

Longitude Arcs @ Equator

$r = 6378137.0\ \mathrm{m}$ WGS-84

1 m arc ⇨ $\Delta\lambda = 1.56786 \times 10^{-7} rad$

$= 8°.98315 \times 10^{-6}$

$= 5'.38989 \times 10^{-4}$

$= 3''.23394 \times 10^{-2}$

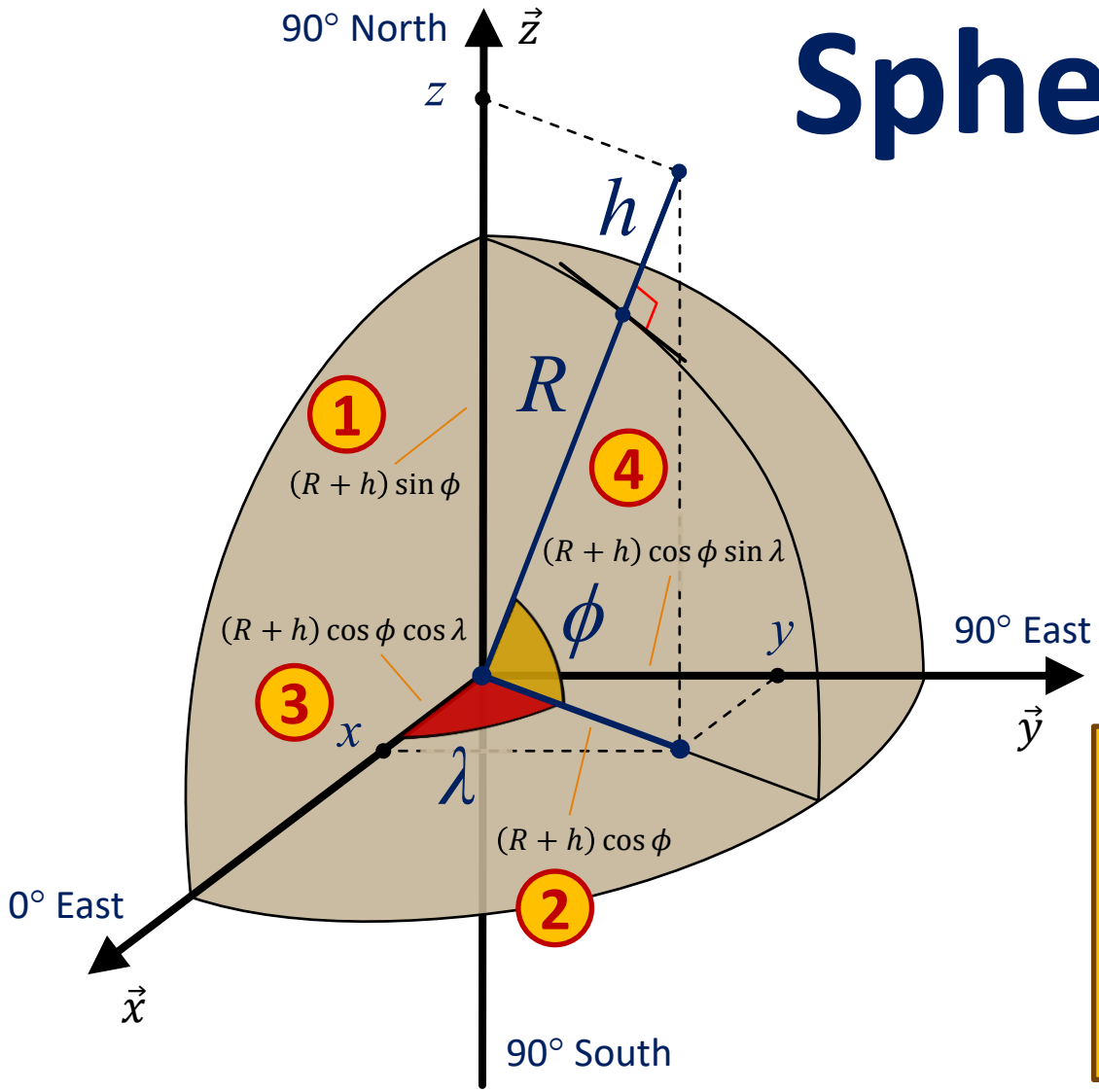
$1\ rad = 6378137.0\ \mathrm{m}$

$1' = 1855.32\ \mathrm{m}$

$1° = 111319\ \mathrm{m}$

$1'' = 30.9221\ \mathrm{m}$

LLH to XYZ on a Spherical World



ϕ - Latitude

Geocentric Latitude

λ - Longitude

h - Height

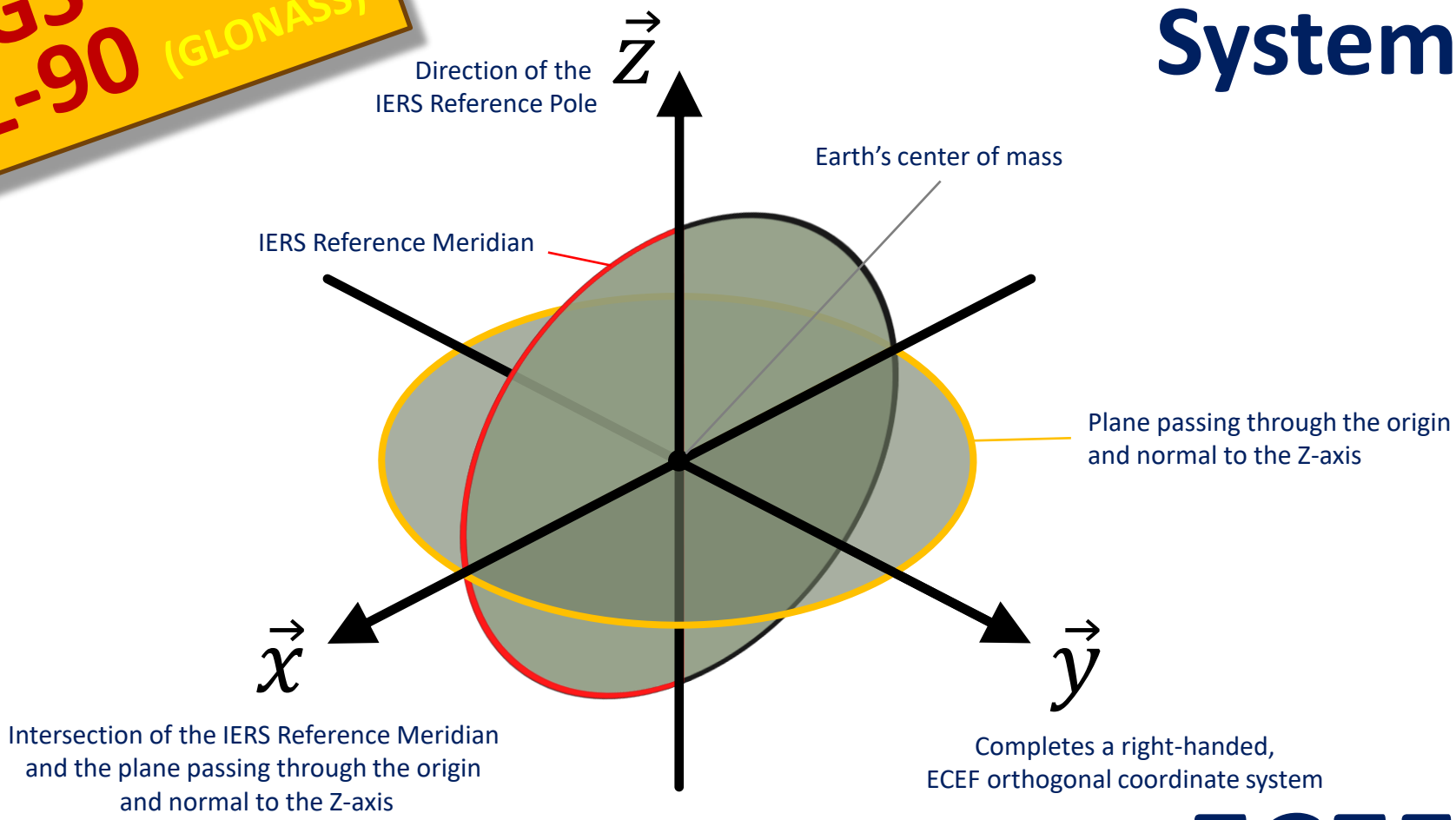
$$x = (R+h)\cos\phi\cos\lambda$$

$$y = (R+h)\cos\phi\sin\lambda$$

$$z = (R+h)\sin\phi$$

Earth-Centered Earth-Fixed Cartesian Coordinate System

WGS-84 (GPS)
PZ-90 (GLONASS)

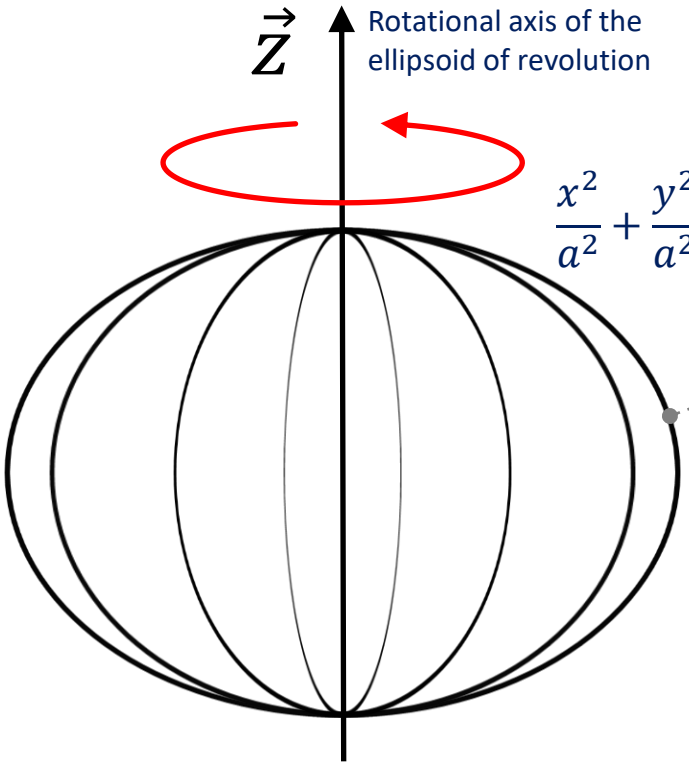


WGS – World Geodetic System
PZ – Parametry Zemli (Russian)
IERS – International Earth Rotation Service

ECEF
Earth-Centered, Earth-Fixed

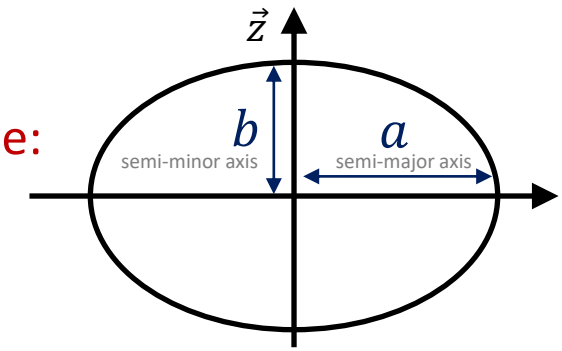
The Shape of the Earth

Ellipsoid of Revolution



$$\frac{x^2}{a^2}+\frac{y^2}{a^2}+\frac{z^2}{b^2}=1$$

ellipse:



Polar ellipsoidal flattening: $f \equiv \frac{a-b}{a}$

(some reference ellipsoids)

		Reference Ellipsoid Name	Semi-major Axis a (m)	Reciprocal of Flattening $1/f$
		Clarke 1880	6378249.145	293.465
Beidou and QZSS	GRS-80	Geodetic Reference System 1980	6378137	298.257222101
	Hayford ellipsoid	International 1924	6378388	297
GLONASS	PZ-90	Parametry Zemli	6378136	298.257839303
GPS	WGS-84	World Geodetic System 1984	6378137	298.257223563

Sources:
National Imagery and Mapping Agency Technical Report NIMA TR8350.2,
Department of Defense World Geodetic System 1984 – Its Definition and Relationships with Local Geodetic Systems, Third Edition, Amendment 1, 3 January 2000, Appendix A.1.
Aeronautical Information Circular of the Russian Federation AIC 04/09, 12.02.2009.

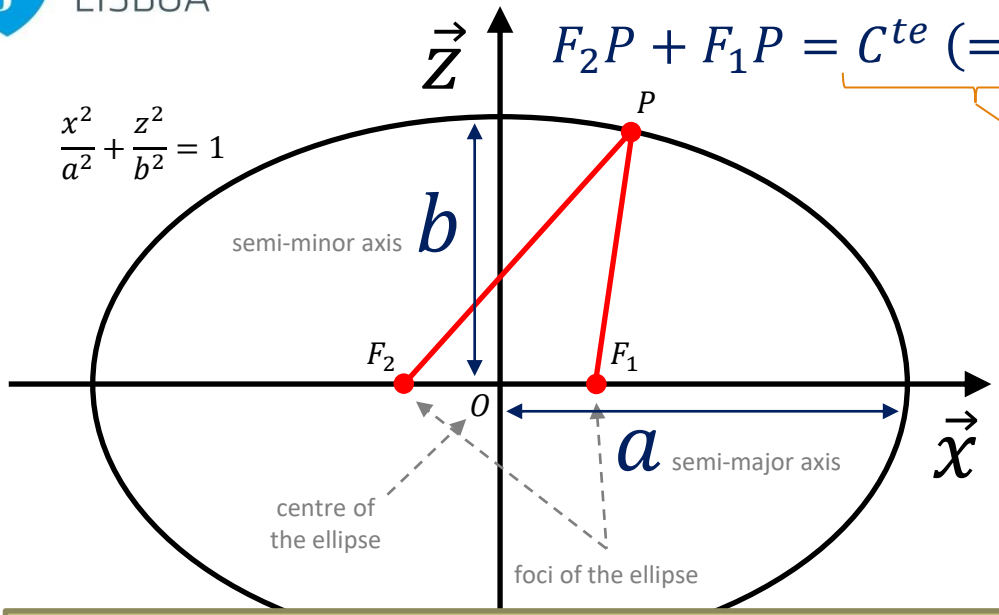
check: Isaac Newton, Principia, Book III, Propositions XVIII and XIX, pp. 405-409, 1687.

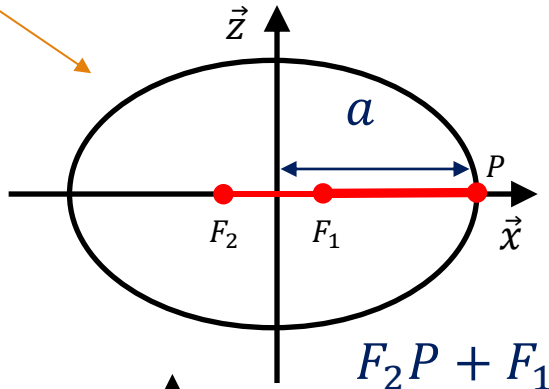
curiosity: Saturn's polar flattening is approx. 1/10

Ellipses

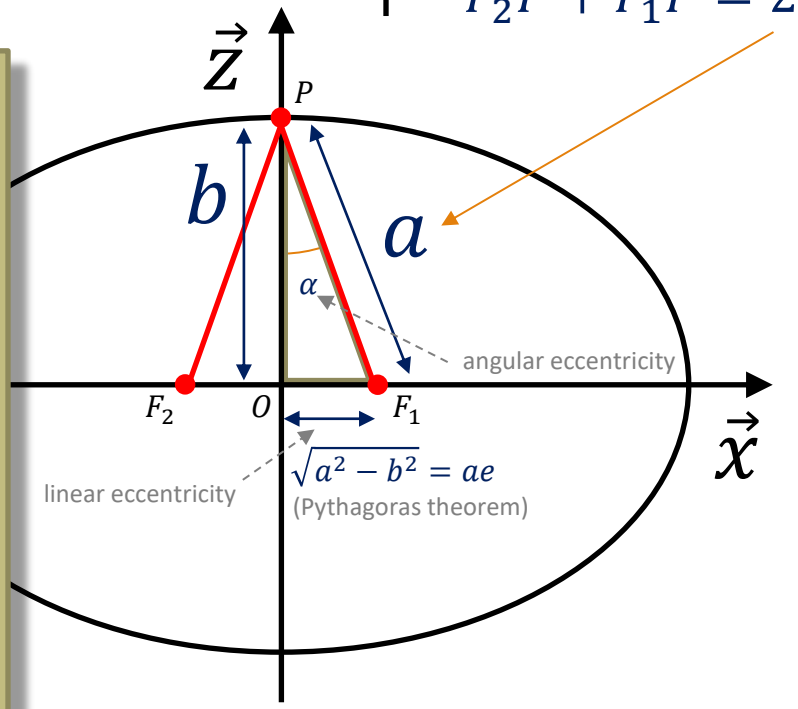
$$F_2P + F_1P = C^{te}\ (= 2a)$$

$$\frac{x^2}{a^2} + \frac{z^2}{b^2} = 1$$





$$F_2P + F_1P = 2a$$



Basic parameters

Semi-major axis: a

Semi-minor axis: b

$$b = a(1 - f)$$

Flattening: $f \equiv \frac{a-b}{a} = 1 - \frac{b}{a}$

First eccentricity: $e \equiv \frac{OF_1}{a} = \frac{\sqrt{a^2-b^2}}{a}$

Second eccentricity: $e' \equiv \frac{OF_1}{b} = \frac{\sqrt{a^2-b^2}}{b} = \frac{e}{1-f}$

$$e^2 = \frac{a^2 - b^2}{a^2} = 1 - \frac{b^2}{a^2} = 1 - (1-f)^2 = 2f - f^2 = \frac{e'^2}{1 + e'^2}$$

$$e'^2 = \frac{a^2 - b^2}{b^2} = \frac{a^2}{b^2} - 1 = \frac{e^2}{1 - e^2}$$

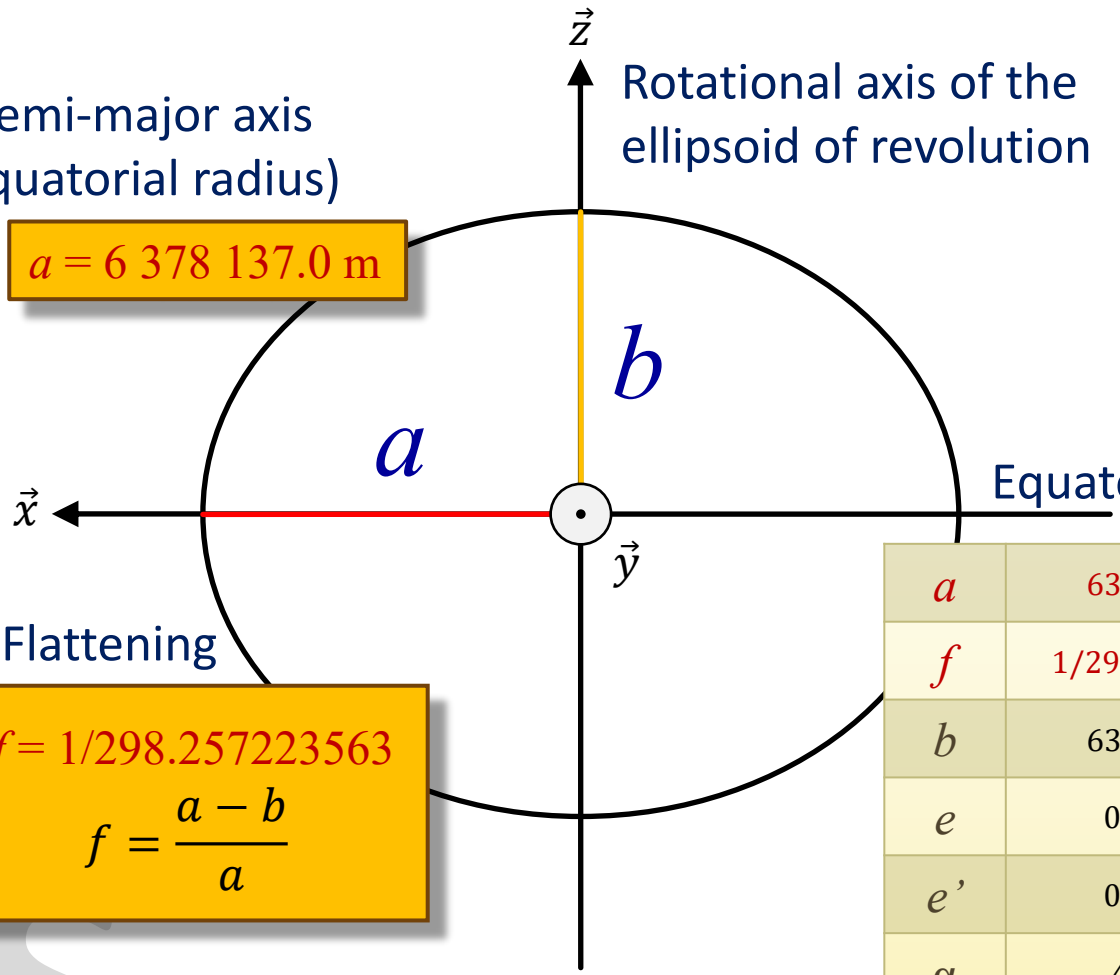
Angular eccentricity (α):

$$\sin\alpha = \frac{OF_1}{a} = e\ ;\ \cos\alpha = \frac{b}{a} = 1 - f\ ;\ \tan\alpha = \frac{OF_1}{b} = e'$$

WGS-84 Ellipsoid

Semi-major axis
(equatorial radius)

$a = 6\,378\,137.0$ m



Rotational axis of the
ellipsoid of revolution

Equatorial plane

Polar Ellipsoidal Flattening

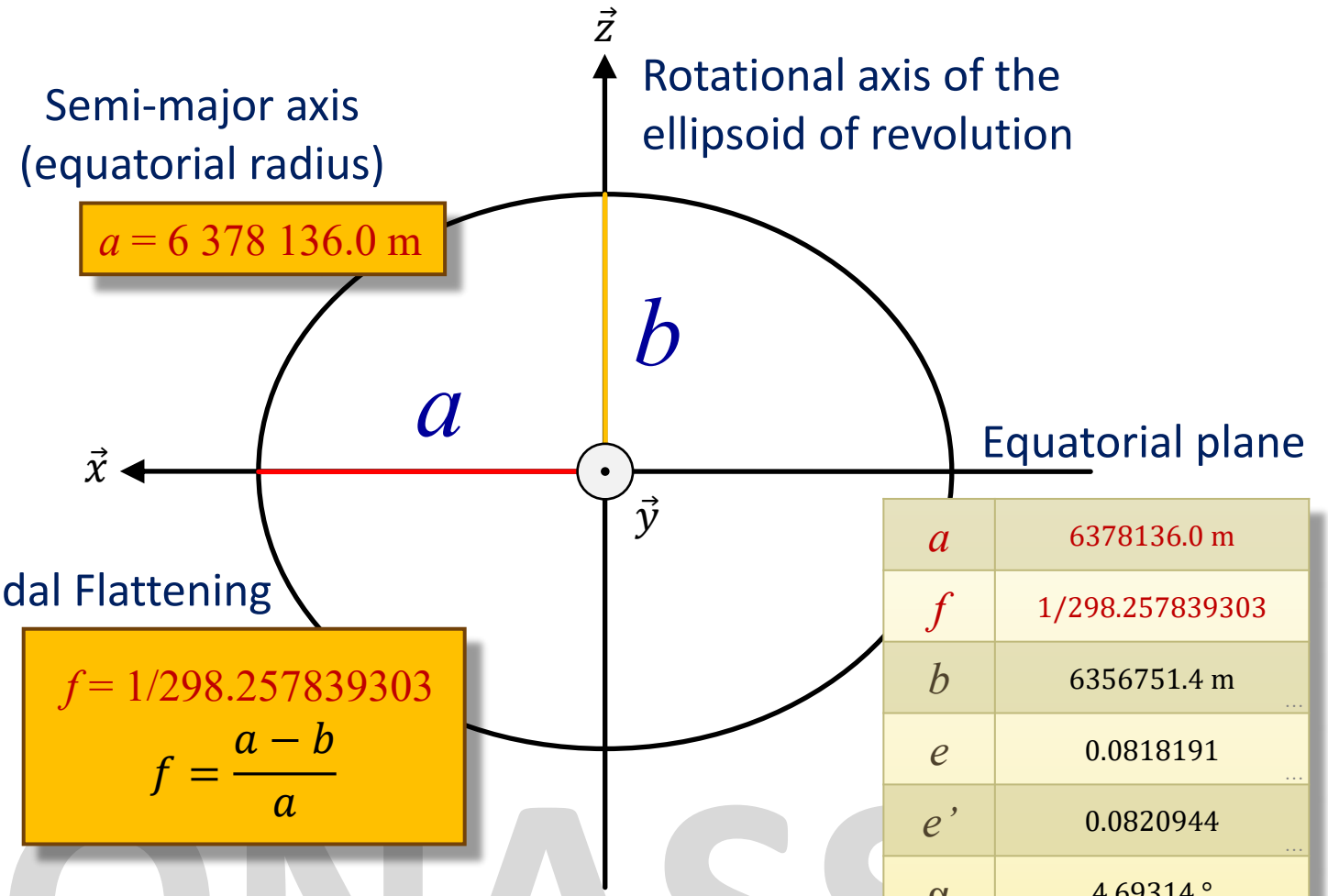
$f = 1/298.257223563$

$$f = \frac{a-b}{a}$$

a	6378137.0 m
f	1/298.257223563
b	6356752.3 m
e	0.0818192
e'	0.0820944
α	4.69314 °
ae	521854 m

GPS

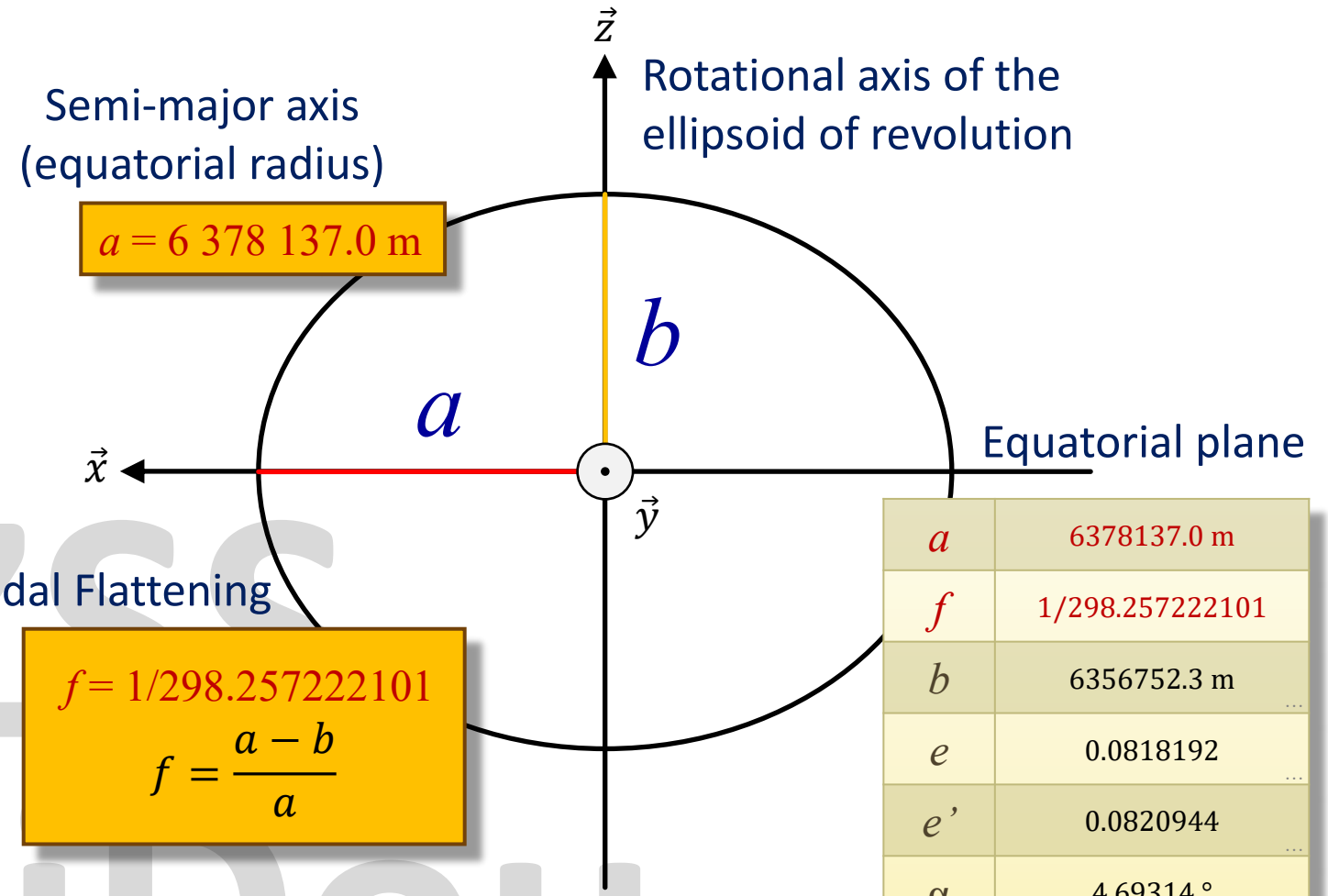
PZ-90 Ellipsoid



a	6378136.0 m
f	1/298.257839303
b	6356751.4 m
e	0.0818191
e'	0.0820944
α	4.69314 °
ae	521853 m

GLONASS

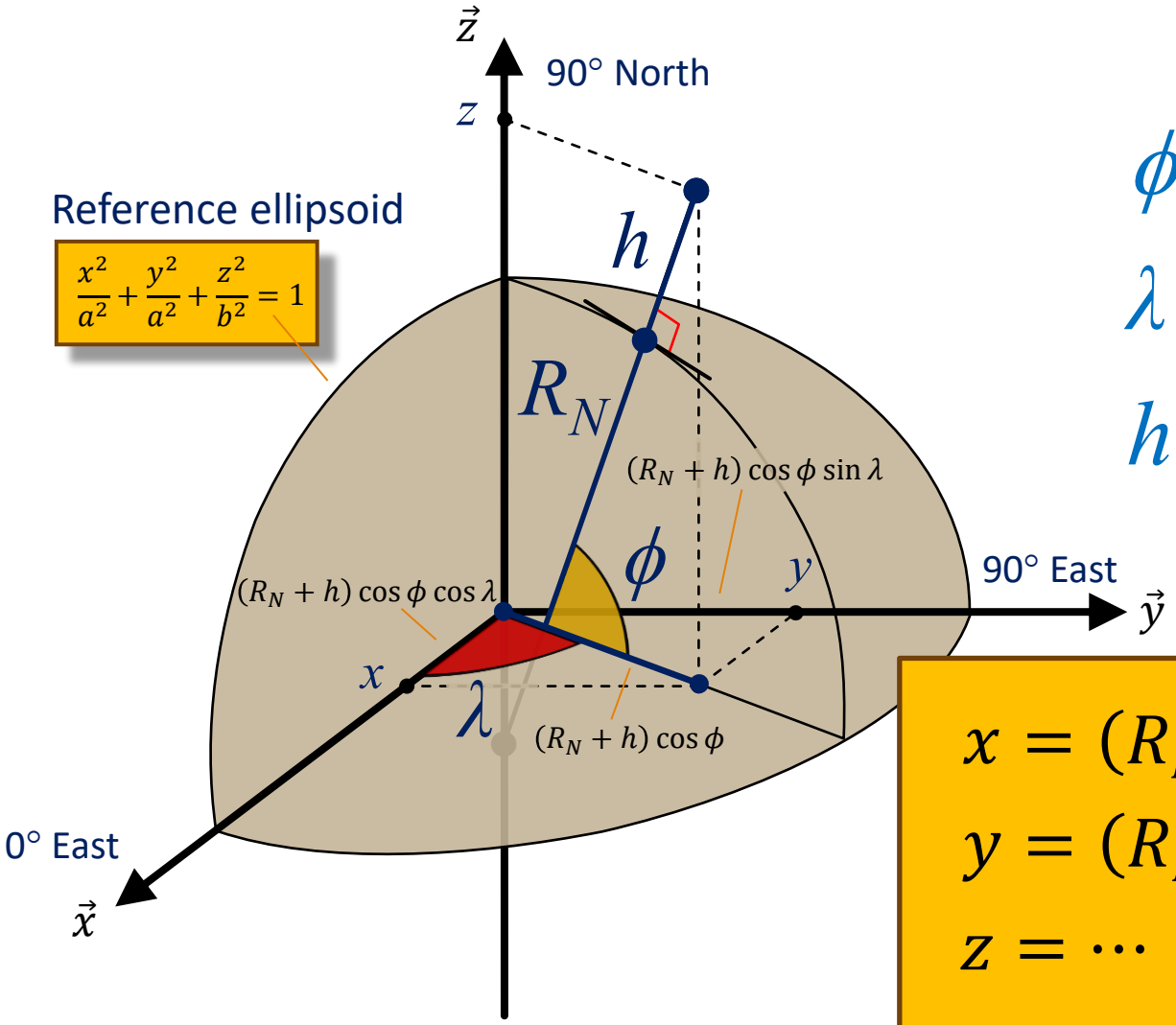
GRS-80 Ellipsoid



a	6378137.0 m
f	1/298.257222101
b	6356752.3 m
e	0.0818192
e'	0.0820944
α	4.69314 °
ae	521854 m

QZSS BeiDou

Geodetic Coordinates



Reference ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{a^2} + \frac{z^2}{b^2} = 1$$

ϕ - Latitude

Geodetic Latitude

λ - Longitude

h - Height

(above the ellipsoid)

$$x = (R_N + h)\cos\phi\cos\lambda$$
$$y = (R_N + h)\cos\phi\sin\lambda$$
$$z = \cdots$$

R_N – Radius of curvature in the prime vertical

LLH_{WGS-84}: [38°44′15.58″ N 9°08′18.67″ W 199.5m]

XYZ_{WGS-84}: [4918526.668m -791212.115m 3969767.140m]

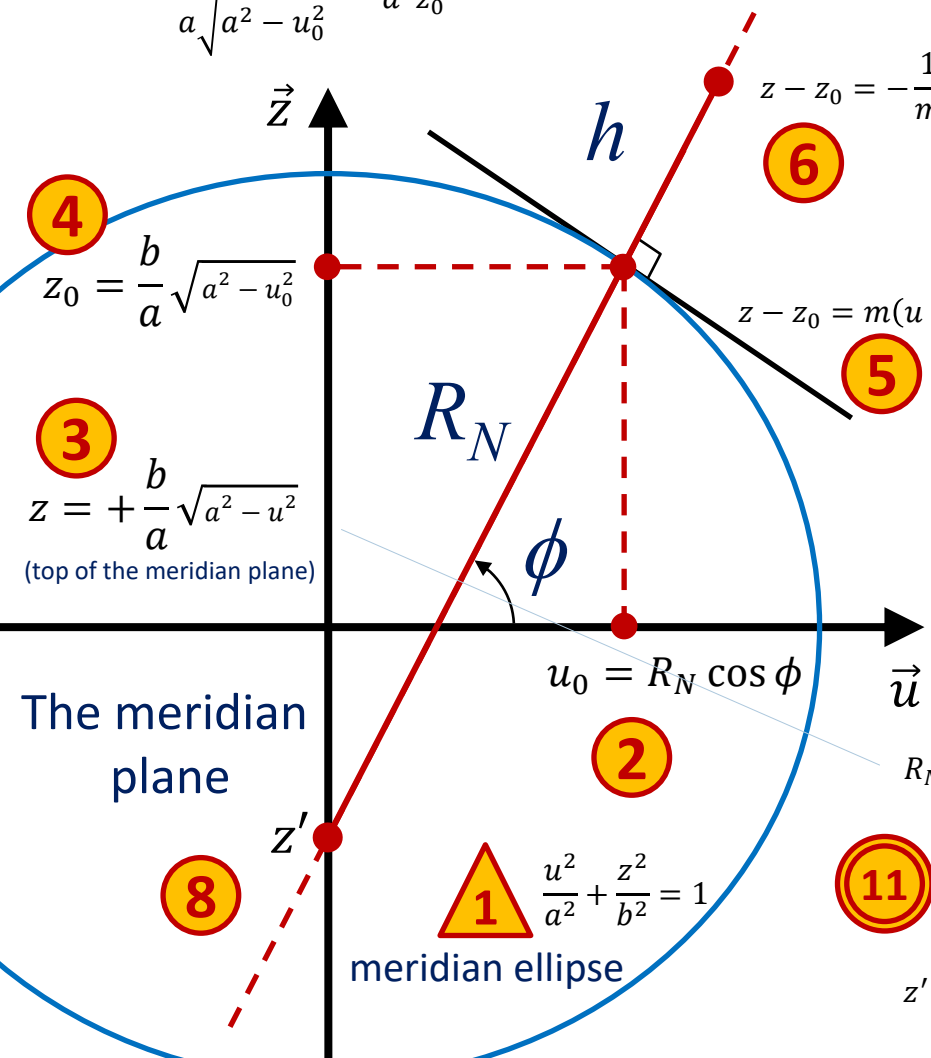
Radius of Curvature in the Prime Vertical

(east-west dividing great circle)

7

$$m = \left.\frac{dz}{du}\right|_{u=u_0} = \left.\frac{b}{a}\frac{1}{2}\left(a^2-u^2\right)^{-1/2}(-2u)\right|_{u=u_0}$$

$$= \frac{-bu_0}{a\sqrt{a^2-u_0^2}} = \frac{-b^2u_0}{a^2z_0}$$



4

$$z_0 = \frac{b}{a}\sqrt{a^2-u_0^2}$$

3

$$z = +\frac{b}{a}\sqrt{a^2-u^2}$$

(top of the meridian plane)

6

$$z - z_0 = -\frac{1}{m}(u-u_0) \xrightarrow[u=0]{9} z' = z_0 + \frac{u_0}{m} = -\left(\frac{a^2}{b^2}-1\right)z_0$$

5

$$z - z_0 = m(u-u_0)$$

2

$$u_0 = R_N\cos\phi$$

1

$$\frac{u^2}{a^2}+\frac{z^2}{b^2}=1$$

8

10

$$R_N^2 = u_0^2 + (z_0-z')^2 = u_0^2 + \frac{a^4}{b^4}z_0^2 = u_0^2 + \frac{a^2}{b^2}\left(a^2-u_0^2\right) =$$

$$= \frac{a^4}{b^2} + \left(1-\frac{a^2}{b^2}\right)R_N^2(\cos\phi)^2 \Leftrightarrow$$

$$\Leftrightarrow R_N^2\left(1-(\cos\phi)^2+\frac{a^2}{b^2}(\cos\phi)^2\right) = \frac{a^4}{b^2} \Leftrightarrow$$

$$\Leftrightarrow R_N = \frac{a}{\sqrt{\frac{b^2}{a^2}(\sin\phi)^2+(\cos\phi)^2}} = \frac{a}{\sqrt{1-\left(1-\frac{b^2}{a^2}\right)(\sin\phi)^2}} \Leftrightarrow$$

$$\boxed{R_N = \frac{a}{\sqrt{1-e^2(\sin\phi)^2}} = \frac{a}{\sqrt{1-f(2-f)(\sin\phi)^2}}}$$

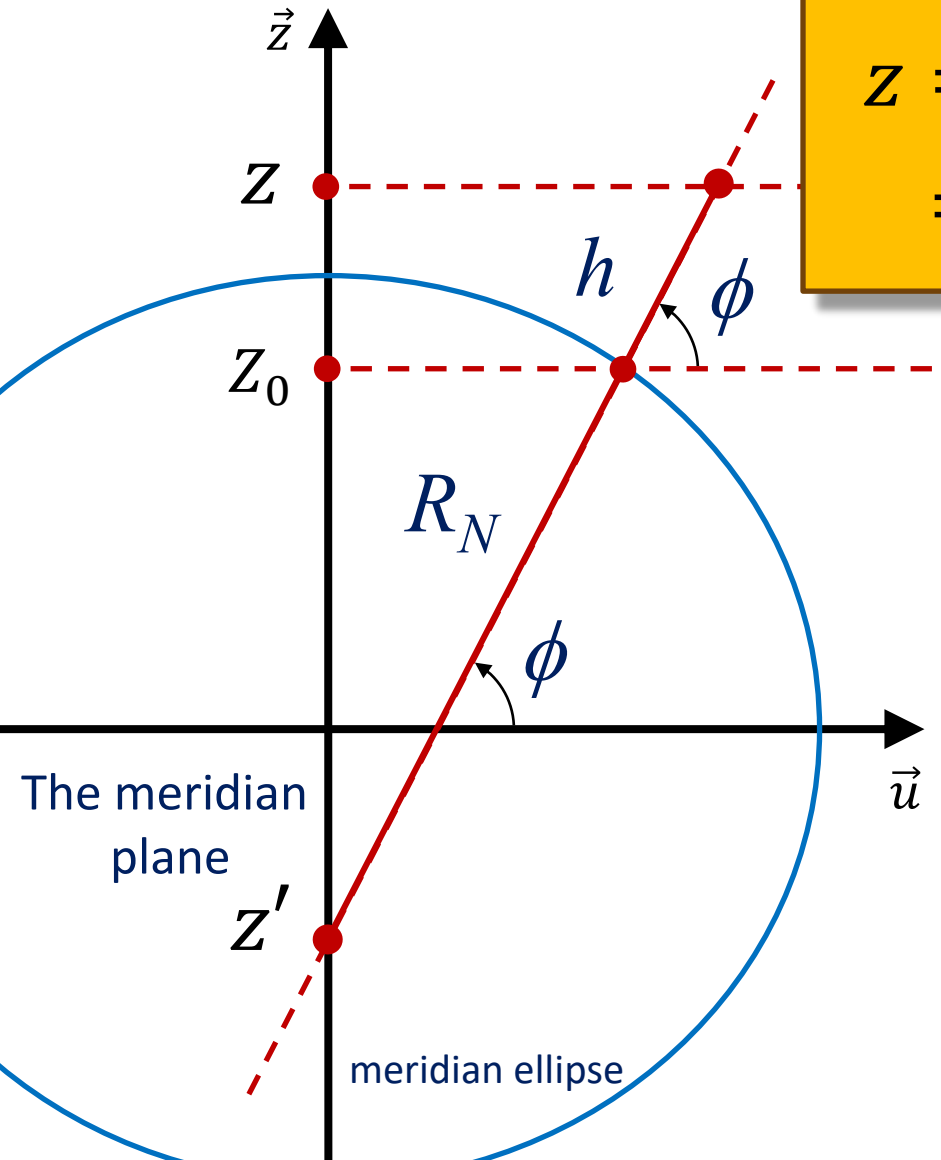
$$R_N\sin\phi = z_0 - z' = \frac{a^2}{b^2}z_0 \Leftrightarrow$$

11

$$\boxed{z_0 = \frac{b^2}{a^2}R_N\sin\phi = \left(1-e^2\right)R_N\sin\phi = (1-f)^2R_N\sin\phi}$$

$$z' = -\left(\frac{a^2}{b^2}-1\right)z_0 \Leftrightarrow \boxed{z' = -e^2R_N\sin\phi = -\left(1-(1-f)^2\right)R_N\sin\phi}$$

Z = …



$$z = z_0 + h \sin\phi$$
$$= \left((1-f)^2 R_N + h\right) \sin\phi$$

$$R_N = \frac{a}{\sqrt{1 - f(2-f)(\sin\phi)^2}}$$

$$z_0 = (1-f)^2 R_N \sin\phi$$

Geodetic to Cartesian Coordinates

LLH to XYZ

$$x = (R_N + h)\cos\phi\cos\lambda$$
$$y = (R_N + h)\cos\phi\sin\lambda$$
$$z = \left((1-f)^2 R_N + h\right)\sin\phi$$

$$R_N = \frac{a}{\sqrt{1 - f(2-f)(\sin\phi)^2}}$$

ϕ – latitude
λ – longitude
h – height (above the ellipsoid)

a – ellipsoid semi-major axis
f – ellipsoidal flattening
R_N – radius of curvature in the prime vertical

Example

Datum: WGS-84

$a = 6\,378\,137.0$ m
$f = 1/298.257223563$

LLH: [38°44'12.52" N 9°08'18.82" W 95.0m]
XYZ: [4918503.838m -791212.210m 3969628.148m]

```
>> xyz=llh2xyz([deg2rad(dms2deg(38,44,12.52)) -deg2rad(dms2deg(9,8,18.82)) 95.0],6378137.0,1/298.257223563)
xyz =
        4918503.83762702          -791212.210237617          3969628.14859058
>>
```

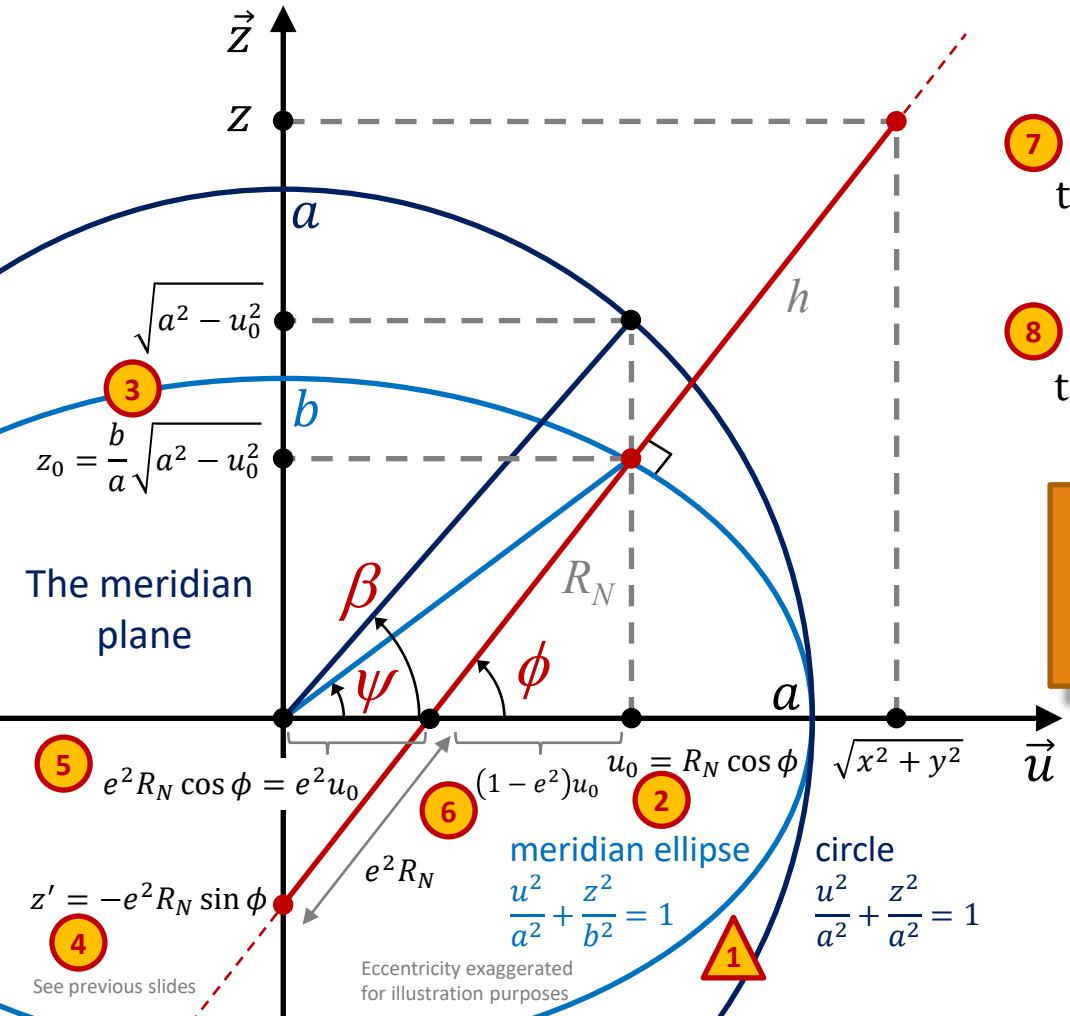
```
function xyz=llh2xyz(llh,a,f)
% LLH2XYZ        Geodetic to cartesian coordinates.
%   xyz=LLH2XYZ(llh,a,f) converts the geodetic coordinates
%   (latitude,longitude,height), in the (1x3) llh row vector,
%   to cartesian coordinates (x,y,z), returned in the (1x3)
%   xyz row vector, based on an ellipsoid with semi-major
%   axis a and flattening f.
%   Parameters a and f are scalar. Multiple n row vectors
%   can be converted if llh is a (nx3) matrix.
%   The following document contain ellipsoidal parameters a
%   and f that can be used in this conversion:
%       Department of Defense World Geodetic System 1984.
%       NIMA TR8350.2. Third Edition. 4 July 1997.
%   Units:
%       lat     radians [-Pi/2,Pi/2]
%       lon     radians [-Pi,Pi]
%       height  meters
%       x,y,z   meters
%       a       meters
%       f       no units

coslat=cos(llh(:,1));
sinlat=sin(llh(:,1));
RN=a./sqrt(1-f*(2-f)*sinlat.^2);

xyz(:,1)=(RN+llh(:,3)).*coslat.*cos(llh(:,2));
xyz(:,2)=(RN+llh(:,3)).*coslat.*sin(llh(:,2));
xyz(:,3)=((1-f)^2*RN+llh(:,3)).*sinlat;

return
```

my MATLAB® toolbox

Latitudes

Geodetic ϕ, Geocentric ψ, Reduced β

a.k.a. eccentric anomaly (when the ellipse is an orbit)



7
$$\tan\phi = \frac{\frac{b}{a}\sqrt{a^2-u_0^2}}{(1-e^2)u_0} = \frac{\frac{a}{b}\sqrt{a^2-u_0^2}}{u_0}$$

8
$$\tan\psi = \frac{\frac{b}{a}\sqrt{a^2-u_0^2}}{u_0}$$

9
$$\tan\beta = \frac{\sqrt{a^2-u_0^2}}{u_0}$$

10
$$\tan\phi = \frac{a}{b}\tan\beta = \frac{a^2}{b^2}\tan\psi$$

$$1-e^2 = \frac{b^2}{a^2}$$

Cartesian (xyz) ↔ Geodetic (ϕλh) Transformations

ϕ - Latitude
(geodetic latitude)
λ - Longitude
h - Height
(above the ellipsoid)

LLH to XYZ

straightforward, direct, exact, without singularities ✓

XYZ to LLH

XY to Lon(λ)

several direct exact solutions (some with singularities)

arctan, Vaníček and Krakiwsky (1982), Zhu (1994), Vermeille (2004)

XY PZ to Lat(ϕ)

direct solutions

exact

approximate

Bowring (1976), Borkowski (1989)

iterative solutions

Hirvonen and Moritz (1963), Bowring (1976), Borkowski (1989)

XY PZ and Lat(ϕ) to H

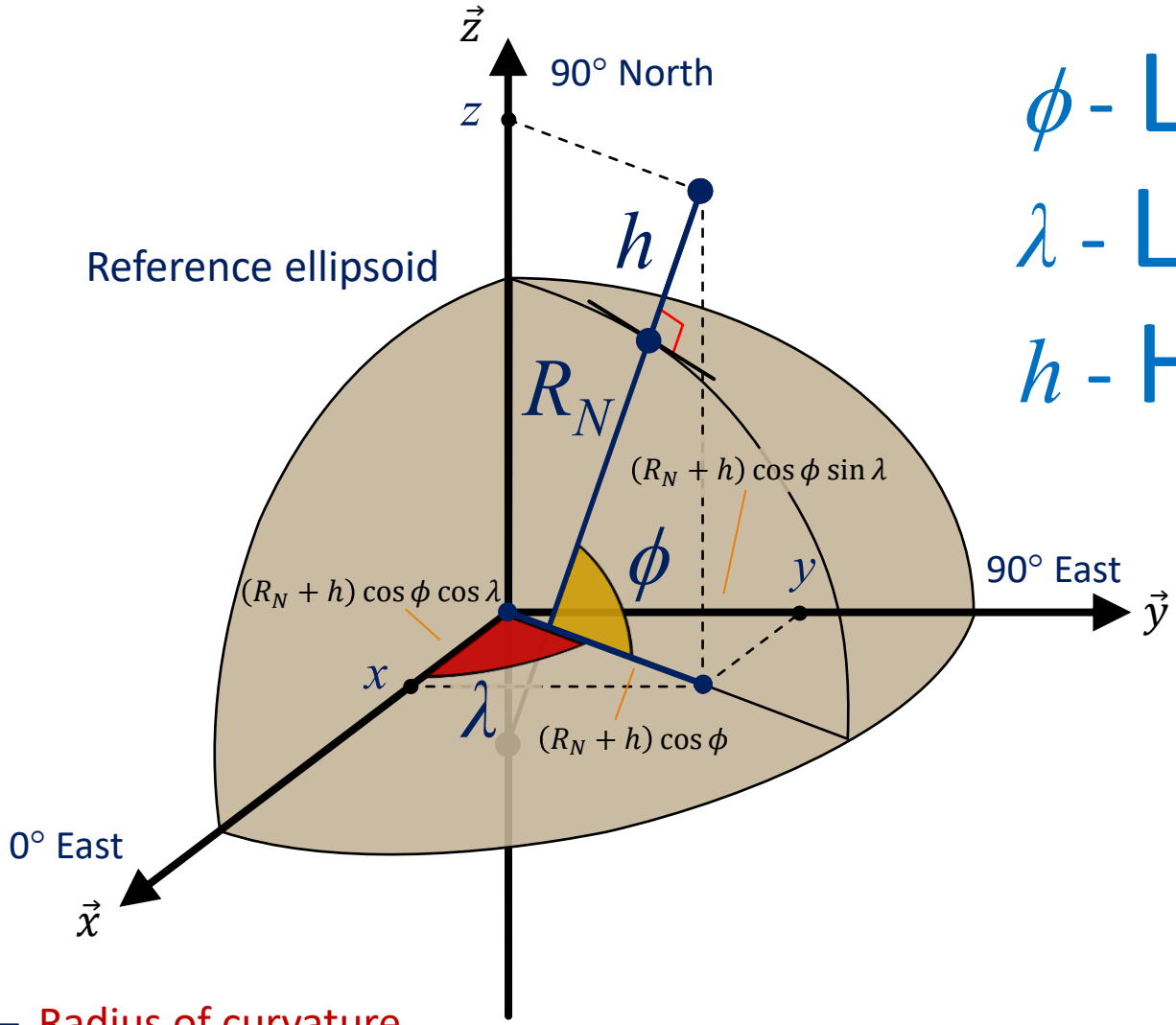
several direct exact solutions (some with singularities)

Hirvonen and Moritz (1963), Bartelme and Meissl (1975), Bowring (1985), Borkowski (1989)

$p = \sqrt{x^2 + y^2}$ (distance from the polar axis)

… just a few of the available methods

Geodetic Coordinates



ϕ - Latitude

Geodetic Latitude

λ - Longitude

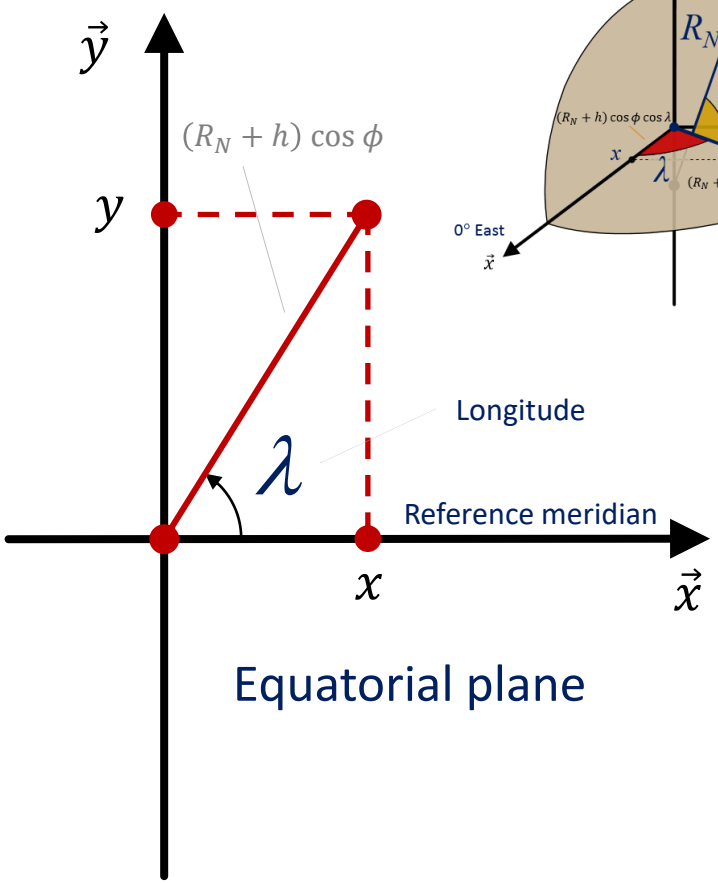
h - Height

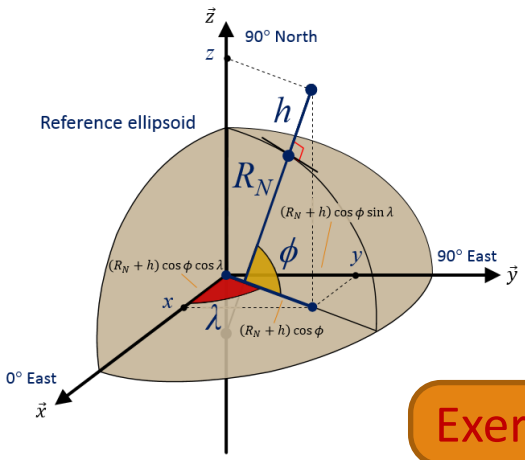
(above the ellipsoid)

R_N – Radius of curvature in the prime vertical

Cartesian to Geodetic Coordinates

XY to Longitude (λ)





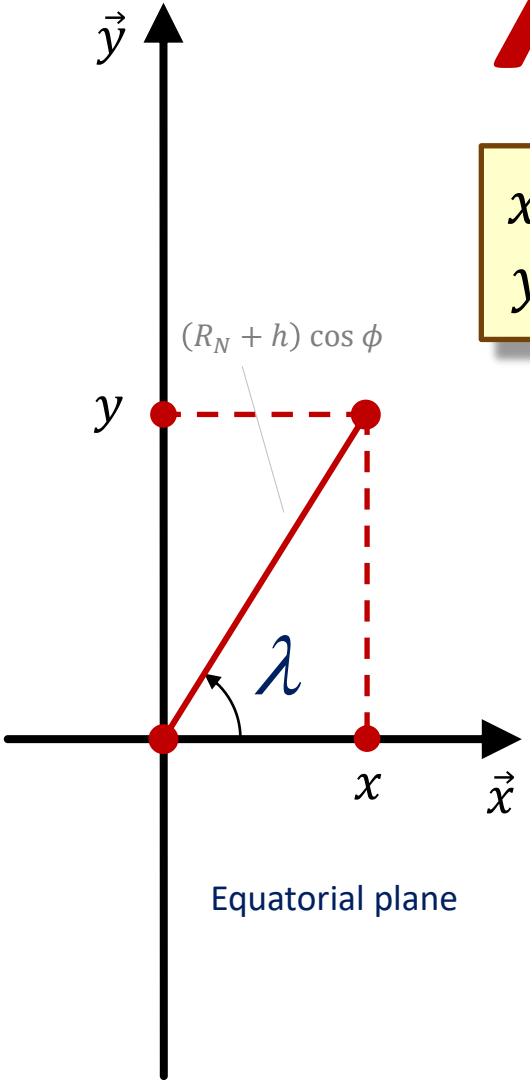
Exercise

Compute the longitude (λ) associated with each of the following Cartesian coordinates:

a) (x,y) = (4358212m, 2516215m)
b) (x,y) = (-2516215m, 4358212m)
c) (x,y) = (-4358212m, -2516215m)
d) (x,y) = (2516215m, -4358212m)
e) (x,y) = (0m, 5032429 m)
f) (x,y) = (0m, -5032429 m)

Cartesian to Geodetic Coordinates

XY to Longitude (λ)



Equatorial plane

$$x = (R_N + h)\cos\phi\cos\lambda$$
$$y = (R_N + h)\cos\phi\sin\lambda$$

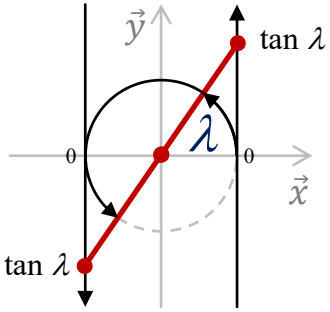
$$\tan\lambda = \frac{y}{x}$$

$\phi \neq \pm\frac{\pi}{2}$

$$\lambda_{[\mathrm{rad}]} = \begin{cases} \arctan\left(\frac{y}{x}\right) & x > 0 & \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[\\ \arctan\left(\frac{y}{x}\right) + \pi & x < 0 \wedge y \geq 0 & \left]\frac{\pi}{2}, \pi\right] \\ \arctan\left(\frac{y}{x}\right) - \pi & x < 0 \wedge y < 0 & \left]-\pi, -\frac{\pi}{2}\right[\\ \frac{\pi}{2} & x = 0 \wedge y > 0 & \\ -\frac{\pi}{2} & x = 0 \wedge y < 0 & \\ \text{undefined} & x = 0 \wedge y = 0 & \end{cases}$$

(range)

Floating point comparison



$\arctan(y/x)$ is in the range $[-\pi/2, \pi/2]$

MATLAB® hint: check `atan2` function

Tangent Half Angle Formulas

$$\lambda = \arctan(\tan\lambda)$$

Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$\lambda = 2\arctan\left(\tan\frac{\lambda}{2}\right)$$

Range: $[-\pi, \pi]$ ✓

$$\tan\frac{\lambda}{2} = \frac{\sin\lambda}{1+\cos\lambda} = \frac{1-\cos\lambda}{\sin\lambda}$$

Cartesian to Geodetic Coordinates

XY to Longitude (λ)

Vaníček and Krakiwsky (1982)

$$\lambda = 2\arctan\left(\frac{y}{\sqrt{x^2+y^2}+x}\right)$$

$\phi \neq \pm\frac{\pi}{2}$

Valid for $x = 0$, but is indeterminate when $x \leq 0$ and $y = 0$ (unusable around $\lambda = \pm\pi$).

$$x = (R_N + h)\cos\phi\cos\lambda$$
$$y = (R_N + h)\cos\phi\sin\lambda$$

$$\sin\lambda = \frac{y}{(R_N + h)\cos\phi}$$

$$\cos\lambda = \frac{x}{(R_N + h)\cos\phi}$$

$$\tan\frac{\lambda}{2} = \frac{\sin\lambda}{1+\cos\lambda}$$

$$(\sin\lambda)^2 + (\cos\lambda)^2 = 1 = \frac{x^2+y^2}{\left((R_N+h)\cos\phi\right)^2} \Leftrightarrow (R_N + h)\cos\phi = \sqrt{x^2+y^2}$$

$\phi \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow \cos\phi \geq 0$

Petr Vaníček, Edward J. Krakiwsky, "Geodesy: The Concepts," Elsevier Science Ltd., Amsterdam, The Netherlands, 1982.

Cartesian to Geodetic Coordinates

XY to Longitude (λ)

Zhu's (1994) Method

$$\lambda = 2\arctan\left(\frac{\sqrt{x^2+y^2}-x}{y}\right)$$

$\phi \neq \pm\frac{\pi}{2}$

Undefined when $y = 0$ (unusable around $\lambda = 0$ or $\lambda = \pm\pi$).

$$x = (R_N + h)\cos\phi\cos\lambda$$
$$y = (R_N + h)\cos\phi\sin\lambda$$

$$\sin\lambda = \frac{y}{(R_N + h)\cos\phi}$$

$$\cos\lambda = \frac{x}{(R_N + h)\cos\phi}$$

$$\tan\frac{\lambda}{2} = \frac{1-\cos\lambda}{\sin\lambda}$$

$$(\sin\lambda)^2 + (\cos\lambda)^2 = 1 = \frac{x^2+y^2}{\left((R_N + h)\cos\phi\right)^2} \Leftrightarrow (R_N + h)\cos\phi = \sqrt{x^2+y^2}$$

$\phi \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow \cos\phi \geq 0$

Jijie Zhu, "Conversion of Earth-Centered Earth Fixed Coordinates to Geodetic Coordinates," IEEE Transactions on Aerospace and Electronic Systems, Vol. 30, No. 3, July 1994.

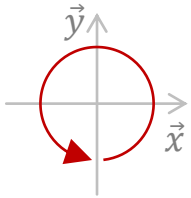
Cartesian to Geodetic Coordinates

XY to Longitude

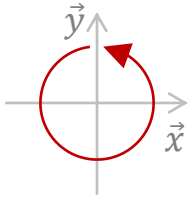
Vermeille's (2004) Method (λ)

$$\lambda_{[rad]} = \begin{cases} \dfrac{\pi}{2} - 2\arctan\left(\dfrac{x}{\sqrt{x^2+y^2}+y}\right), & y \geq 0 \\ -\dfrac{\pi}{2} + 2\arctan\left(\dfrac{x}{\sqrt{x^2+y^2}-y}\right), & y < 0 \end{cases}$$

λ is continuous from $-\pi/2$ to $3\pi/2$



λ is continuous from $-3\pi/2$ to $\pi/2$



Hugues Vermeille, "Computing Geodetic Coordinates from Geocentric Coordinates," Journal of Geodesy, Vol. 78, pp. 94-95, 2004.

Cartesian to Geodetic Coordinates

XY to Longitude

Vermeille's (2004) Method (λ)

Proof:

$$\lambda_{[rad]} = 2\arctan\left(\frac{y}{\sqrt{x^2+y^2}+x}\right)$$

Vaníček and Krakiwsky (1982)

$$y \geq 0 \quad \lambda_{[rad]} = \frac{\pi}{2} - 2\arctan 1 + 2\arctan\left(\frac{y}{\sqrt{x^2+y^2}+x}\right) = \frac{\pi}{2} - 2\arctan\left(\frac{\sqrt{x^2+y^2}+x-y}{\sqrt{x^2+y^2}+x+y}\right)$$

$$= \frac{\pi}{2} - 2\arctan\left(\left(\frac{x+\sqrt{x^2+y^2}-y}{\sqrt{x^2+y^2}+x+y}\right)\left(\frac{\sqrt{x^2+y^2}+y}{x}\right)\left(\frac{x}{\sqrt{x^2+y^2}+y}\right)\right) \qquad \left(\lambda = \frac{\pi}{2}, x = 0\right)$$

$$= \frac{\pi}{2} - 2\arctan\left(\frac{x}{\sqrt{x^2+y^2}+y}\right)$$

$$y < 0 \quad \lambda_{[rad]} = -\frac{\pi}{2} + 2\arctan 1 + 2\arctan\left(\frac{y}{\sqrt{x^2+y^2}+x}\right) = -\frac{\pi}{2} + 2\arctan\left(\frac{\sqrt{x^2+y^2}+x+y}{\sqrt{x^2+y^2}+x-y}\right)$$

$$= -\frac{\pi}{2} + 2\arctan\left(\left(\frac{x+\sqrt{x^2+y^2}+y}{\sqrt{x^2+y^2}+x-y}\right)\left(\frac{\sqrt{x^2+y^2}-y}{x}\right)\left(\frac{x}{\sqrt{x^2+y^2}-y}\right)\right) \qquad \left(\lambda = -\frac{\pi}{2}, x = 0\right)$$

$$= -\frac{\pi}{2} + 2\arctan\left(\frac{x}{\sqrt{x^2+y^2}-y}\right)$$

$$\arctan(-u) = -\arctan(u)$$

$$\arctan u + \arctan v = \arctan\left(\frac{u+v}{1-uv}\right)$$

Hugues Vermeille, "Computing Geodetic Coordinates from Geocentric Coordinates," Journal of Geodesy, Vol. 78, pp. 94-95, 2004.

Cartesian to Geodetic Coordinates

$$\tan\lambda = \frac{y}{x}$$

$$\phi = \arctan\left(\frac{z + e'^2 b(\sin\beta)^3}{p - e^2 a(\cos\beta)^3}\right)$$

$h =$ (use Hirvonen and Moritz - 1963)

XYZ to LLH

Bowring (1976)

approximate solution

(but, for $-5000m \leq h \leq 10000m$ the maximum error in ϕ is 0.000000030")

$$R_N(\phi) = \frac{a}{\sqrt{1 - f(2-f)(\sin\phi)^2}}$$

$$b = a(1-f) \qquad p = \sqrt{x^2 + y^2}$$

$$e^2 = 1 - \frac{b^2}{a^2} \qquad e'^2 = \frac{a^2}{b^2} - 1$$

$$\beta = \arctan\left(\frac{a}{b}\frac{z}{p}\right)$$

ϕ – latitude (geodetic)
λ – longitude
h – height (above the ellipsoid)
a – ellipsoid semi-major axis
b – ellipsoid semi-minor axis
f – ellipsoidal flattening
R_N – radius of curvature in the prime vertical

Bowring (1985)

$$h = p\cos\phi + z\sin\phi - \frac{a^2}{R_N(\phi)}$$

B.R. Bowring, "Transformation from Spatial to Geographical Coordinates," Survey Review, Vol. XXIII, No. 181, 1976, pp. 323-327.

B.R. Bowring, "The Accuracy of Geodetic Latitude and Height Equations," Survey Review, Vol. XXVIII, No. 218, 1985, pp. 202-206.

Reino Antero Hirvonen, Helmut Moritz, "Practical Computation of Gravity at High Altitudes," Report No. 27, Institute of Geodesy, Photogrammetry and Cartography, The Ohio State University, Columbus, Ohio, May 1963, pp. 3-4.

Example

Datum: WGS-84

$a = 6\,378\,137.0$ m
$f = 1/298.257223563$

```
XYZ:  [4918503.838m -791212.210m 3969628.148m]
LLH:  [38°44´12.52´´N 9°08´18.82´´W 95.0m]
```

Cartesian to Geodetic Coordinates

XYZ to LLH

Heikkinen (1982)

$$\tan\lambda = \frac{y}{x}$$

$$F = 54b^2z^2$$

$$G = r^2 + (1-e^2)z^2 - e^2(a^2-b^2)$$

$$c = e^4Fr^2/G^3$$

$$s = \sqrt[3]{1 + c + \sqrt{c^2+2c}}$$

$$P = \frac{F}{3(s+(1/s)+1)^2G^2}$$

$$Q = \sqrt{1+2e^4P}$$

$$r_0 = -\frac{Pe^2r}{1+Q} + \sqrt{\frac{a^2}{2}\left(1+\frac{1}{Q}\right) - \frac{P(1-e^2)z^2}{Q(1+Q)} - \frac{Pr^2}{2}}$$

$$U = \sqrt{(r-e^2r_0)^2 + z^2}$$

$$V = \sqrt{(r-e^2r_0)^2 + (1-e^2)z^2}$$

$$z_0 = \frac{b^2z}{aV}$$

$$h = U\left(1 - \frac{b^2}{aV}\right)$$

$$\phi = \arctan\left(\frac{z+e'^2z_0}{r}\right)$$

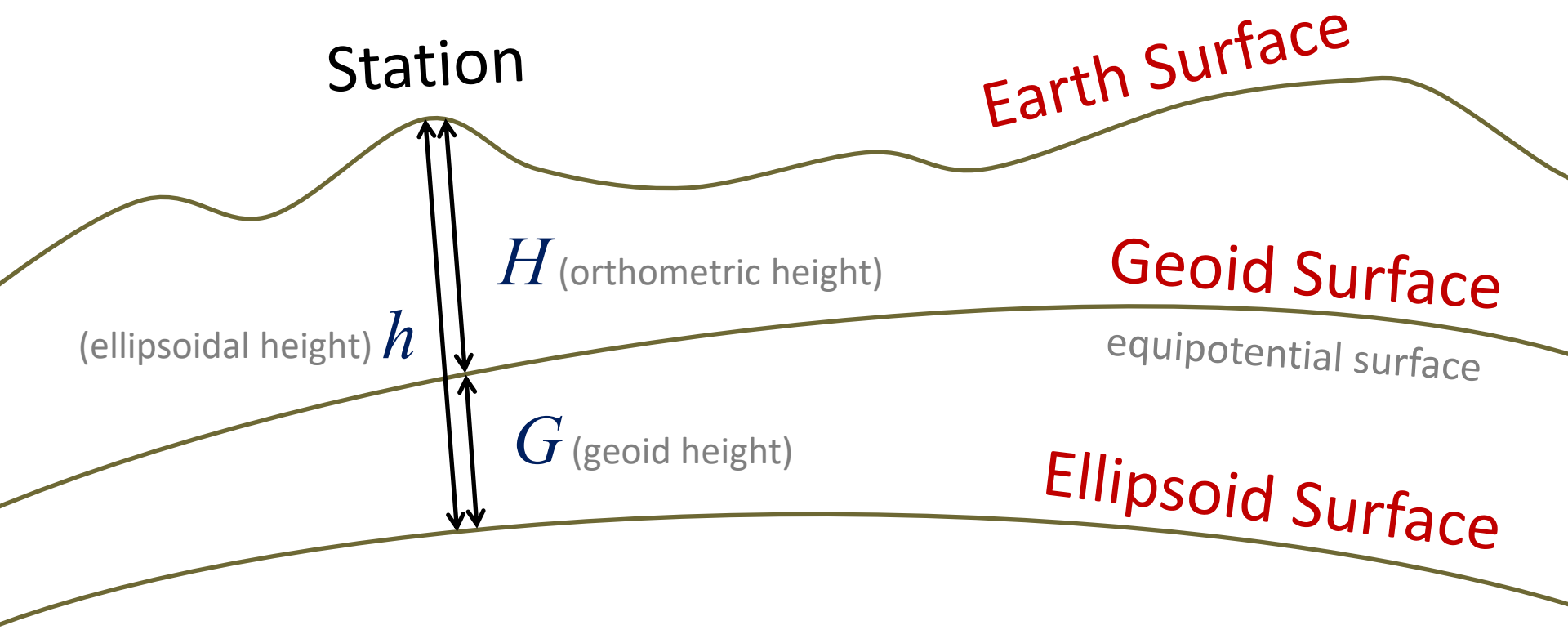
exact solution

$$b = a(1-f) \qquad r = \sqrt{x^2+y^2}$$

$$e^2 = 1 - \frac{b^2}{a^2} \qquad e'^2 = \frac{a^2}{b^2} - 1$$

ϕ – latitude
λ – longitude
h – height (above the ellipsoid)
a – ellipsoid semi-major axis
b – ellipsoid semi-minor axis
f – ellipsoidal flattening

Markku Heikkinen, “Geschlossene Formeln zur Berechnung Räumlicher Geodätischer Koordinaten aus Rechtwinkligen Koordinaten,“ Zeitschrift für Vermessungswesen,107 Jahrgang, 1982, S. 207-211.

The Shape of the Earth

The Geoid